

SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (AUTONOMOUS)

(Approved by AICTE, New Delhi, Affiliated to JNTUK, Kakinada)

Accredited by NAAC with 'A+' Grade.

Recognised as Scientific and Industrial Research Organisation

SRKR MARG, CHINA AMIRAM, BHIMAVARAM – 534204 W.G.Dt., A.P., INDIA

Regulation: R23		II / IV - B.Tech. I - Semester							
CIVIL ENGINEERING									
COURSE STRUCTURE (With effect from 2023-24 admitted Batch onwards)									
Course Code	Course Name	Category	L	T	P	Cr	C.I.E.	S.E.E.	Total Marks
B23BS2104	Numerical Techniques and Statistical Methods	BS	3	0	0	3	30	70	100
B23HS2101	Universal Human Values -II: Understanding Harmony and Ethical Human Conduct	HS	2	1	0	3	30	70	100
B23CE2101	Surveying	PC	3	0	0	3	30	70	100
B23CE2102	Strength of Materials	PC	3	0	0	3	30	70	100
B23CE2103	Fluid Mechanics	PC	3	0	0	3	30	70	100
B23CE2104	Surveying Field Work	PC	0	0	3	1.5	30	70	100
B23CE2105	Strength of Materials Lab	PC	0	0	3	1.5	30	70	100
B23CE2106	Building Planning and Drawing	SEC	0	1	2	2	30	70	100
B23MC2104	Building Materials and Construction	MC	2	0	0	-	30	-	30
TOTAL			16	2	8	20	270	560	830

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
B23BS2104	BS	3	--	--	3	30	70	3 Hrs.
NUMERICAL TECHNIQUES AND STATISTICAL METHODS								
(For CE)								
Course Objectives: Students are expected to								
1.	Elucidate the different numerical methods to solve nonlinear algebraic equations							
2.	Disseminate the use of different numerical techniques for carrying out numerical integration and Solutions of ODE with initial conditions							
3.	Analyze various statistical measures of a few discrete and continuous probability distributions.							
4.	Develop a framework for testing of hypothesis in giving inferences about Population parameters							
Course Outcomes:At the end of this course, Students will be able to								
S.No	Outcome							Knowledge Level
1.	Apply Numerical methods to find an approximate root of an algebraic or transcendental equations.							K3
2.	Predict appropriate interpolation technique to interpolate or extrapolate a given data with equal and unequal intervals.							K3
3.	Solve Ordinary differential equation with initial conditions (IVP's) by using different Numerical techniques.							K3
4.	Solve real time problems using discrete and continuous probability distributions.							K3
5.	Apply appropriate tests of hypothesis to draw the inferences on Parameters based on Statistics and the maximum error of the estimate using the concepts of Sampling theory.							K3
SYLLABUS								
UNIT-I (10Hrs)	Algebraic and Transcendental Equations: Introduction, Bisection method, Secant method, Method of false position, Iteration method, and Newton-Raphson method.							
UNIT-II (10 Hrs)	Interpolation: Newton's forward and backward formulae for interpolation, Gauss forward and backward interpolation, Interpolation with unequal intervals – Lagrange's interpolation formula.							
UNIT-III (10 Hrs)	Numerical integration, Solution of ordinary differential equations with initial conditions: Trapezoidal rule– Simpson's $1/3^{\text{rd}}$ and $3/8^{\text{th}}$ rule– Solution of initial value problems by Taylor's series– Picard's method of successive approximations– Euler's method –Runge-Kutta method (second and fourth order) – Milne's Predictor and Corrector Method.							

UNIT-IV (10 Hrs)	Probability and Distributions Baye's theorem – Random variables – Discrete and Continuous random variables – Distribution functions – Probability mass function, Probability density function and Cumulative distribution functions – Mathematical Expectation and Variance – Binomial, Poisson, Uniform and Normal distributions.
UNIT-V (10 Hrs)	Sampling Theory Introduction – Population and Sample, Sampling distribution of Means and Variance (definition only) – Point and Interval estimations – Maximum error of estimate – Central limit theorem (without proof). Testing of Hypothesis : Null and Alternative Hypothesis, Type I and Type II errors, Level of significance, One tail and two-tailed tests, Tests of significance for large samples (single proportion and difference of proportions), Small Samples: t-test for single and difference of means, F-test, χ^2 -test for goodness of fit.
Textbooks:	
1.	B. S. Grewal , Higher Engineering Mathematics, 44 th Edition, Khanna Publishers.
2.	Miller and Freund's , Probability and Statistics for Engineers, 7/e, Pearson, 2008.
Reference Books:	
1.	Steven C. Chapra , Applied Numerical Methods with MATLAB for Engineering and Science, Tata Mc. Graw Hill Education.
2.	M. K. Jain, S.R.K. Iyengar and R.K. Jain , Numerical Methods for Scientific and Engineering Computation, New Age International Publications.
3.	Lawrence Turyn , Advanced Engineering Mathematics, CRC Press.
4.	S. C. Gupta and V.K. Kapoor , Fundamentals of Mathematical Statistics, 11/e, Sultan Chand & Sons Publications, 2012.
5.	Shron L. Myers, Keying Ye, Ronald E Walpole , Probability and Statistics Engineers and the Scientists, 8 th Edition, Pearson 2007.
6.	Jay I. Devore , Probability and Statistics for Engineering and the Sciences, 8 th Edition, Cengage
e-Resources	
1.	https://archive.nptel.ac.in/courses/127/106/127106019/
2.	https://archive.nptel.ac.in/courses/111/105/111105077/

Course Code	Category	L	T	P	C	CIE	SEE	Exam
B23HS2101	HS	2	1	--	3	30	70	3 Hrs.
UNIVERSAL HUMAN VALUES-II: UNDERSTANDING HARMONY AND ETHICAL HUMAN CONDUCT								
(Common to all Programmes of Engineering)								
Course Objectives: The objective of this course is to make the student aware of								
1	Essential complementarity between 'Values' and 'Skills' to ensure sustained happiness and prosperity which are the core aspirations of all human beings.							
2	Harmony in the human being, family, society and nature/existence							
3	Holistic perspective towards life, profession and happiness							
Course Outcomes: At the end of this course student will be able to								
S. No.	Outcome							Knowledge Level
1	Explain the role of value education in achieving basic human aspirations.							K2
2	Summarize needs to obtain harmony in self(I).							K2
3	Describe criteria for human-human relationship and harmony in society							K2
4	Explain four orders of nature and our existence							K2
5	Interpret significance of harmony in holistic development							K2
SYLLABUS								
UNIT-I (9 Hrs)	Introduction to Value Education: Understanding Value Education- Need, Basic Guidelines, Content and Process for Value Education Purpose and motivation for the course. Self-exploration as the Process for Value Education - Sharing about Oneself. Myers-Briggs Type Indicator (MBTI) Personality Test. Continuous Happiness and Prosperity – the Basic Human Aspirations and their Fulfilment. Right Understanding, Relationship and Physical Facility (Holistic Development and the Role of Education) - Exploring Human Consciousness. Happiness and Prosperity – Current Scenario. Method to Fulfil the Basic Human Aspirations - Exploring Natural Acceptance-understanding and living in harmony at various levels.							
UNIT-II (9 Hrs)	Harmony in the Human Being: Understanding Human being as the Co-existence of the Self and the Body. Distinguishing between the Needs of the Self and the Body - Exploring the difference of Needs of Self (I) and Body (Happiness and Physical Facility).							

	The Body as an Instrument of the Self (I)' (I being the doer, seer and enjoyer). Understanding Harmony in the Self(I) - Exploring Sources of Imagination in the Self(I). Harmony of the Self (I) with the Body (characteristics and activities of 'I' and harmony in 'I'). Programme to ensure self-regulation(<i>Sanyam</i>) and Health(<i>Swasth</i>)- Exploring Harmony of Self (I) with the Body.
UNIT-III (9 Hrs)	Harmony in the Family and Society: Harmony in the Family – the Basic Unit of Human Interaction. 'Trust' – the Foundational Value in Relationship- Exploring the Feeling of Trust- intention and competence. 'Respect' – as the Right Evaluation- Exploring the Feeling of Respect. Other Feelings, Justice in Human-to-Human Relationship. Understanding Harmony in the Society- (society being an extension of family). Vision for the Universal Human Order- Exploring Systems to fulfil Human Goal.
UNIT-IV (6 Hrs)	Harmony in the Nature/Existence: Understanding the harmony in the Nature. Interconnectedness and mutual fulfillment among the four orders of nature- recyclability and self-regulation in nature. Realizing Existence as Co-existence at All Levels - Understanding Existence as Co-existence of mutually interacting units in all pervasive space. Holistic perception of harmony at all levels of existence.
UNIT-V (9 Hrs)	Implications of the Holistic Understanding – a Look at Professional Ethics: Natural acceptance of human values. Definitiveness of (ethical) human conduct. A Basis for Humanistic Education, Humanistic Constitution and Humanistic Universal Order. Competence in professional ethics: a. Ability to utilize the professional competence for augmenting universal human order b. Ability to identify the scope and characteristics of people friendly and eco-friendly production systems, c. Ability to identify and develop appropriate technologies and management patterns for above production systems. Holistic technologies, production systems and management models- typical case studies. Strategies for transition towards value based life and profession (from the present state to Universal Human Order): a. At the level of individual: as socially and ecologically responsible engineers, technologists and managers, b. At the level of society: as mutually enriching institutions and organizations.
Text Books	
1.	R R Gaur, R Sangal, G P Bagaria. "Human Values and Professional Ethics", Excel Books, New Delhi, 2010
2.	R R Gaur, R Asthana, G P Bagaria. "Teachers' Manual for A Foundation Course in Human Values and Professional Ethics", 2nd Revised Edition, Excel Books, New Delhi, 2019. ISBN

	978-93-87034-53-2
Reference Books:	
1.	Jeevan Vidya: Ek Parichaya, A Nagaraj, Jeevan Vidya Prakashan, Amarkantak, 1999.
2.	Human Values, A.N. Tripathi, New Age Intl. Publishers, New Delhi, 2004.
3.	The Story of Stuff (Book).
4.	The Story of My Experiments with Truth - by Mohandas Karamchand Gandhi
5.	Small is Beautiful - E. F Schumacher
6.	Slow is Beautiful - Cecile Andrews
7.	Economy of Permanence - J C Kumarappa
8.	Bharat Mein Angreji Raj – Pandit Sunderlal
9.	Rediscovering India - by Dharampal
10.	Hind Swaraj or Indian Home Rule - by Mohandas K. Gandhi
11.	India Wins Freedom - Maulana Abdul Kalam Azad
12.	Vivekananda - Romain Rolland (English)
e-Resources	
1.	https://fdp-si.aicte-india.org/UHV%20II%20Teaching%20Material/UHV%20II%20Lecture%2023-25%20Ethics%20v1.pdf
2.	https://fdp-si.aicte-india.org/UHV-II%20Class%20Note.php
3.	https://fdp-si.aicte-india.org/download/FDPTeachingMaterial/3-days%20FDP-SI%20UHV%20Teaching%20Material/Day%203%20Handouts/UHV%203D%20D3-S2A%20Und%20Nature-Existence.pdf

Estd. 1980

ENGINEERING COLLEGE
AUTONOMOUS

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
B23CE2101	PC	3	--	--	3	30	70	3 Hrs.
SURVEYING								
(For CE)								
Course Objectives: The objectives of this course is to make the student aware of								
1.	Principle and methods of surveying							
2.	Measuring horizontal and vertical-distances and angles. Perform calculations based on the observation.							
3.	Identifying source of errors and rectification methods							
4.	Surveying principles to determine areas and volumes and setting out curves							
5.	Modern surveying equipment's for accurate results							
Course Outcomes: At the end of the course students will be able to								
S.No	Outcome							Knowledge Level
1.	Apply Principles and methods of surveying for measuring the distances and angles using instruments.							K3
2.	Determine the levels for the contouring along with areas and volumes.							K3
3.	Determine the working principles of Theodolite and measurement of horizontal and vertical angles along with identifying source of errors and rectification methods.							K3
4.	Determine the principles of tacheometry and setting of curves.							K3
5.	Use of modern surveying techniques and instruments for accurate results.							K3
SYLLABUS								
UNIT-I (10Hrs)	Introduction and Basic Concepts: Introduction, Objectives, classification and principles of surveying, Surveying accessories. Introduction to Compass, leveling and Plane table surveying. Linear distances- Approximate methods, Direct Methods- Chains- Tapes, ranging, Tape corrections. Prismatic Compass- Bearings, included angles, Local Attraction, Magnetic Declination, and dip –systems and W.C.B and Q.B systems of locating bearings.							
UNIT-II (10 Hrs)	Leveling- Types of levels, methods of levelling, and Determination of levels, Effect of Curvature of Earth and Refraction. Contouring- Characteristics and uses of Contours, methods of contour surveying. Areas - Determination of areas consisting of irregular boundary and regular boundary. Volumes -Determination of volume of earth work in cutting and embankments for level							

	section, capacity of reservoirs.
UNIT-III (10 Hrs)	Theodolite Surveying: Types of Theodolites, temporary adjustments, measurement of horizontal angle by repetition method and reiteration method, measurement of vertical Angle, Trigonometrical leveling when base is accessible and inaccessible. Traversing: Methods of traversing, traverse computations and adjustments, Introduction to Omitted measurements
UNIT-IV (10 Hrs)	Tachometric Surveying: Principles of Tachometry, stadia and tangential methods of Tachometry Curves: Types of curves and their necessity, elements of simple, compound, reverse curves
UNIT-V (10 Hrs)	Modern Surveying Methods: Principle and types of E.D.M. Instruments, Total station- advantages and Applications. Introduction to Global Positioning System. Introduction to Drone survey and LiDAR Survey (Light Detection And Ranging). Photogrammetry Surveying: Introduction, Basic concepts, perspective geometry of aerial photograph, relief and tilt displacements, terrestrial photogrammetry, flight planning; Stereoscopy, Photographic mapping- mapping using paper prints, mapping using stereo-plotting instruments, mosaics, map substitutes.
Textbooks:	
1.	Surveying (Vol –1 & 2) by Duggal S K, Tata McGraw Hill Publishing Co. Ltd. New Delhi, 5 th edition, 2019
2.	Textbook of Surveying by C Venkatramaiah , Universities Press 1 st Edition, 2011
Reference Books:	
1.	Surveying (Vol – 1), by B. C. Punmia, Ashok Kumar Jain and Arun Kumar Jain - Laxmi Publications (P) ltd., New Delhi, 18 th edition 2024
2.	Surveying (Vol – 3), by B. C. Punmia, Ashok Kumar Jain and Arun Kumar Jain - Laxmi Publications (P) ltd., New Delhi 16 th 2023
3.	Plane Surveying and Higher Surveying by Chandra A M, New age International Pvt. Ltd., Publishers, New Delhi, 3 rd Edition, 2015
4.	Surveying and Levelling by N. BasakTata McGraw Hill Publishing Co. Ltd. New Delhi, 4 th edition, 2014
5.	Surveying (Vol 1, 2 & 3), by Arora K R, Standard Book House, Delhi. Edition: 12th, 2015
e-Resources	
1.	https://nptel.ac.in/courses/105107122
2.	https://archive.nptel.ac.in/courses/105/104/105104101/

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
B23CE2102	PC	3	--	--	3	30	70	3 Hrs.
STRENGTH OF MATERIALS								
(For CE)								
Course Objectives: The objectives of this course is to make the student aware of								
1.	The concepts of stress, strain and elastic constants and their relations, and to impart knowledge on types of stresses in structural elements for better understanding of behavior of solid materials.							
2.	Shear force, bending moments and deflection of beams.							
3.	The calculations of crippling loads in columns and struts.							
Course Outcomes: At the end of the course students will be able to								
S.No	Outcome							Knowledge Level
1.	Predict the behavior of a material under the influence of different external loading conditions.							K3
2.	Plot the Shear Force (SF) and Bending Moment (BM) diagrams for cantilever and simply supported beams under different loading conditions.							K3
3.	Calculate the bending and shear stresses in beam sections.							K3
4.	Determine the principal stresses and strains using mathematical methods and engineering principles and apply the basic methods to find slope and deflection of determinate beams.							K3
5.	Determine the critical loads for columns to assess their stability against buckling and calculate the torsional stresses in shafts.							K3
SYLLABUS								
UNIT-I (10Hrs)	Stresses and strains: Definitions of stress and strain – types of stresses and strains – Elasticity – Hooke’s law –Stress – Strain diagram for Mild steel – working stress- factor of safety- Lateral strain –Poisson’s ratio and volumetric strain – Elastic Moduli and the relationship between them –Bars of varying section– Composite bars.							
UNIT-II (10 Hrs)	Shear Force & Bending Moment Diagrams: Definition of beam – Types of beams – concept of SF and BM – SF &BM diagrams for cantilever and simply supported beams subjected to point loads, UDL, uniformly varying loads, couples and combination of these loads–Point of contraflexure–Relationship b/w S.F, B.M and rate of loading.							
UNIT-III (10 Hrs)	Flexural Stresses: Theory of simple Bending – Assumptions–Derivation of Bending Equation - Neutral axis – Determination of bending stresses–section modulus of rectangular &Circular sections (Solid and Hollow), I, T, channel sections – Problems on							

	Design of simple beam sections. Shear Stresses: Derivation of shear stress formula – shear stress distribution across various beam sections like rectangular, circular, Triangular, I, T, angle sections, built up beams.
UNIT-IV (10 Hrs)	Principal Stresses and Strains: Introduction-stresses on an inclined section of a bar under axial loading- compound stresses-Normal and tangential stresses on an inclined plane for biaxial stresses-Two perpendicular normal stresses accompanied by a state of simple shear-Mohr's circle of stress-Principal planes and principal stresses- Construction of Mohr's Circle (graphical Method) Deflections of Beams: Deflections in (i) Cantilever and (ii) Simply supported beams using (a) Double integration and (b) Macaulay's method
UNIT-V (10 Hrs)	Columns & Struts: Introduction– axially loaded compression members – crushing load– Buckling load (or) crippling load – Euler's theory for long columns – Assumptions – Derivations of Euler's critical load formula - slenderness ratio – Rankine formula for both long and short columns- Problems on Euler's and Rankine's formula. Torsion of Circular Shafts: Theory of pure Torsion – Torsional moment of Resistance – polar section Modulus - Derivation of Torsional equation –Problems on circular shafts only.
Textbooks:	
1.	Bansal, R. K., Strength of Materials, Laxmi Publications, 1996.
2.	Rajput, R.K., Strength of Materials, S. Chand & Co., Ltd., 1996.
Reference Books:	
1.	Timoshenko, S.P. and Gere, J.M., Mechanics of Materials, Tata McGraw Hill, 1992.
2.	Ramamrutham, S. and Narayanan, R., Strength of Materials, Dhanpat Rai Publishing company, 2020.
3.	Vazirani. and Ratwani, N.M. Strength of Materials, Vol. I, Kanna Publishers, 1996
4.	Gunneswararao, T. D. and Andal Mudim by, Strength of Materials: Fundamentals and Applications, Cambridge University Press, 2018.
e-Resources	
1.	https://archive.nptel.ac.in/courses/105/105/105105108/
2.	https://onlinecourses.nptel.ac.in/noc23_me140/preview

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
B23CE2103	PC	3	--	--	3	30	70	3 Hrs.
FLUID MECHANICS								
(For CE)								
Course Objectives: The objectives of this course is to make the student aware of								
1.	The basic properties of fluids.							
2.	The characteristics and behavior of fluids under static, kinematic, and dynamic conditions.							
3.	The analysis of fluid flow through pipes.							
Course Outcomes: At the end of the course students will be able to								
S.No	Outcome							Knowledge Level
1.	Compute the various properties of fluids.							K3
2.	Apply the laws of fluid statics.							K3
3.	Apply the concept of fluid kinematics to fluids flowing under atmospheric pressure.							K3
4.	Apply the Principles of equation of motion to flowing fluids under forces.							K3
5.	Determine the losses and fluid discharge in the pipes.							K3
SYLLABUS								
UNIT-I (10Hrs)	Basic concepts and definitions: Distinction between a fluid and a solid; Density, Specific weight, Specific gravity, Kinematic and dynamic viscosity; Variation of viscosity with temperature, Newton law of viscosity; Vapor pressure, Boiling point, Surface tension, Capillarity, Bulk modulus of elasticity, Compressibility							
UNIT-II (10 Hrs)	Fluid statics: Fluid Pressure: Pressure at a point, Pascal's law, pressure variation with temperature, density and altitude. Piezometer, U-Tube Manometer, Single Column Manometer, U Tube Differential Manometer. Pressure gauges, Hydrostatic pressure and force: horizontal, vertical and inclined surfaces. Buoyancy and stability of floating bodies.							
UNIT-III (10 Hrs)	Fluid kinematics: Classification of fluid flow: steady and unsteady flow; uniform and non-uniform flow; laminar and turbulent flow; rotational and irrotational flow; compressible and incompressible flow; ideal and real fluid flow; one, two and three-dimensional flows; Streamline, path line, streak line and stream tube; stream function, velocity potential function. One, two and three-dimensional continuity equations in Cartesian coordinates.							

UNIT-IV (10 Hrs)	Fluid Dynamics: Surface and body forces; Equations of motion - Euler's equation; Bernoulli's equation – Derivation; Energy Principle; Practical applications of Bernoulli's equation: Venturimeter, orifice meter and Pitot tube; Momentum principle; Forces exerted by fluid flow on pipe bend; Vortex Flow – Free and Forced; Definitions of Reynolds Number, Froude Number, Mach Number, Weber Number and Euler Number.
UNIT-V (10 Hrs)	Analysis Of Pipe Flow: Energy losses in pipelines; Darcy – Weisbach equation; Minor losses in pipelines; Hydraulic Grade Line and Total Energy Line; Concept of equivalent length – Pipes in Parallel and Series.
Textbooks:	
1.	P. M. Modi and S. M. Seth, Hydraulics and Fluid Mechanics, Standard Book House 22 nd , 2019.
2.	K. Subrahmanya, Theory and Applications of Fluid Mechanics, Tata McGraw Hill, 2 nd edition 2018.
Reference Books:	
1.	R. K. Bansal, A text of Fluid mechanics and hydraulic machines, Laxmi Publications (P) Ltd., New Delhi 11 th edition, 2024.
2.	N. Narayana Pillai, Principles of Fluid Mechanics and Fluid Machines, Universities Press Pvt Ltd, Hyderabad. 3rd Edition 2009.
3.	Fluid Mechanics by Frank M. White, Henry Xue, Tata McGraw Hill, 9 th edition, 2022.
4.	C. S. P. Ojha, R. Berndtsson and P. N. Chadramouli, Fluid Mechanics and Machinery, Oxford University Press, 2010.
5.	Introduction to Fluid Mechanics & Fluid Machines by S K Som, Gautam Biswas, S Chakraborty Tata McGraw Hill, 3rd edition 2011.
e-Resources	
1.	https://nptel.ac.in/courses/112104118
2.	https://onlinecourses.nptel.ac.in/noc22_ce85/preview

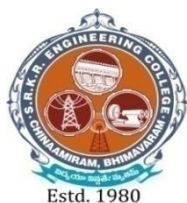
Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
B23CE2104	PC	--	--	3	1.5	30	70	3 Hrs
SURVEYING FIELD WORK								
(For CE)								
Course Objectives: The objective of this course is to make the student aware of								
1	The basic methods of Chain Surveying and Compass Surveying.							
2	Determining the area and volume by interpreting the data obtained from surveying activities.							
3	Drafting field notes from survey data.							
4	Modern equipment such as total station.							
Course Outcomes: At the end of the course students be able to								
S.No	Outcome							Knowledge Level
1	Perform chain surveying to find the linear measurements in simple boundaries							K3
2	Perform the compass surveying to determine the direction of any line using compass surveying							K3
3	Perform the fly levelling surveys to find the levels.							K3
4	Perform the theodolite and tacheometric surveys to find linear and angular measurements							K3
5	Perform the survey by using the modern equipment like total station to find areas, contouring by interpreting the data from surveying activities.							K3
LIST OF FIELDS WORKS								
Students have to perform any 8 of the following Experiments:								
1	Study of chains and its accessories, Aligning, Ranging and Chaining and Chain Traversing							
2	Compass Traversing– Measuring Bearings & arriving included angles							
3	Levelling: Longitudinal & Cross section levelling for an open traverse and closed traverse							
4	Theodolite: Distance between two in-accessible points by Theodolite							
5	Trigonometric Leveling-finding height of the building.							
6	Determination of Heights and distance using Principles of tacheometric surveying							
7	Setting out a simple circular curve							
8	Determine area using total station							
9	Contouring using total station							
10	Stakeout using total station							
11	Distance, Gradient, Different height between two inaccessible points using total station.							
Reference Books:								
1	Arthur R Benton and Philip J Taety, Elements of Plane Surveying, Mc Graw Hill							
2	Surveying and levelling by R. Subramanian, Oxford university press, New Delhi							
3	Arora KR “SurveyingVol1,2&3), Standard Book House, Delhi							

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
B23CE2105	PC	--	--	3	1.5	30	70	3 Hrs.
STRENGTH OF MATERIALS LAB								
(For CE)								
Course Objectives:								
This course focuses on teaching students how to conduct tests on different materials to determine their mechanical properties. They will learn techniques such as stretching materials, bending beams, twisting, checking hardness, and compressing materials. Students will practice safe data collection and compliance with protocols. They will also develop skills in analyzing test results and documenting their findings in detailed reports.								
Course Outcomes: At the end of this course student will be able to								
S.No	Outcome							Knowledge Level
1	Conduct tensile strength test and draw stress-strain diagrams for ductile metals							K3
2	Perform bending test and determine load-deflection curve of steel/wood							K3
3	Conduct torsion test and determine torsion parameters							K3
4	Perform hardness, impact and shear strength tests and calculate hardness numbers, impact and shear strengths							K3
5	Conduct tests on closely coiled and open coiled springs and calculate deflections							K3
SYLLABUS								
Students have to perform the following Experiments:								
1	Tension test on Mild/HYSD bars							
2	Bending test on (Steel/Wood) Cantilever beam.							
3	Bending test on simply supported beam							
4	Hardness test							
5	Test on springs							
6	Compression test on wood/concrete/bricks							
7	Izod / Charpy Impact test on metals							
8	Shear test on metals							
9	Verification of Maxwell's reciprocal theorem							
Reference Books:								
1	Hemant Sood, Laboratory Manual on Testing of Engineering Materials, New Age International Publishers, 2013							
2	C.B.Kukreja, K. Kishore, Ravi Chawala, Material Testing Laboratory Manual, Standard Publishers Distributors, 2001							

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
B23CE2106	SEC	--	1	2	2	30	70	3 Hrs.
BUILDING PLANNING AND DRAWING								
(For CE)								
Course Objectives: The objective of this course is to make the student aware of								
1	Different building bye-laws and regulations							
2	Planning aspects of residential buildings and public buildings							
3	Training exercises on various signs and bonds							
4	Training exercises on different building units							
5	The skills and methods of planning of various buildings.							
Course Outcomes: At the end of this course student will be able to								
S.No	Outcome							Knowledge Level
1	Plan various buildings as per the building by-laws							K3
2	Distinguish the relation between the plan, elevation and cross section and identify the form and functions among the buildings							K3
3	Draw signs and bonds							K3
4	Draw different building units							K3
5	Learn the skills of drawing building elements and plan the buildings as per requirements							K3
SYLLABUS								
Students have to perform any 8 of the following Experiments:								
1	Detailing & Drawing of Sign Conventions.							
2	Detailing & Drawing of English bond.							
3	Detailing & Drawing of Flemish bond.							
4	Detailing & Drawing of Doors.							
5	Detailing & Drawing of Windows.							
6	Detailing & Drawing of Ventilators & Roofs.							
7	Drawing of Line Diagram of Residential Buildings by using Building Bye- Laws.							
8	Drawing of Plan, Elevation & Section from line diagram for a single Storey Building.							
9	Drawing of Plan, Elevation & Section for Hospital Building.							
10	Drawing of Plan, Elevation & Section for Industrial Building.							
Reference Books:								
1	Building planning and drawing by M. Chakraborti.							
2	Building drawing, M G Shah, C M Kale and S Y Patki, Tata McGraw Hill, New Delhi							
3	Civil Engineering drawing and House planning, B. P. Verma, Khanna publishers, NewDelhi							

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
B23MC2104	PC	2	--	--	--	30	--	3 Hrs.
BUILDING MATERIALS AND CONSTRUCTION								
(For CE)								
Course Objectives: The objectives of this course is to make the student aware of								
1.	various building materials and assess their structural suitability for construction.							
2.	The importance of cement and lime in construction processes, focusing on their significance in enhancing structural integrity and durability.							
3.	The properties and applications of sustainable and smart building materials for innovative, eco-friendly construction.							
Course Outcomes: At the end of the course students will be able to								
S.No	Outcome							Knowledge Level
1.	Interpret the quality of stones, bricks and clay products.							K2
2.	Explain function of various materials like wood, glass, paints and building components.							K2
3.	Interpret the properties of lime, cement, and aggregates and their use in concrete production.							K2
4.	Illustrate the importance of masonry, finishing and form woks.							K2
5.	Explain various sustainable and smart building materials							K2
SYLLABUS								
UNIT-I (10Hrs)	Building Stones, Bricks and Tiles Stone-Building stones, classification of building stones, quarrying procedures, characteristics of good building stone. Bricks-Composition of brick earth, manufacturing of brick, characteristics of good brick, field and lab test. Tiles -Types of tiles, manufacturing of tiles, structural requirements of tiles.							
UNIT-II (8Hrs)	Wood, Glass, Paints Wood- structure, types of wood, properties of wood, seasoning, defects, alternative material for wood. Glass-types of glasses, manufacturing of glass. Paints -Constituents of paints, types of paints.							
UNIT-III (12Hrs)	Lime , Cement and Aggregates Lime: Various ingredients of lime – Constituents of lime stone – Classification of lime – various methods of manufacture of lime.							

	Cement: Portland cement- Chemical Composition – Hydration, setting and fineness of cement. Various types of cement and their properties. Various field and laboratory tests for Cement Aggregates Classification of aggregate – Coarse and fine aggregates- particle shape and texture – Bond and Strength of aggregate – Specific gravity – Bulk Density, porosity and absorption – Moisture content of Aggregate- Bulking of Sand
UNIT-IV (8Hrs)	Masonry and Finishing, Form Works Brick Masonry- Types and bonds. Stone Masonry- Types. Finishing- plastering, pointing and cladding- Types of ACP (Aluminum composite panel). Form Works - requirements, standards, Scaffolding, shoring and under pinning.
UNIT-V (10Hrs)	Sustainable and Green Advanced Building Materials Introduction –Advanced building materials, Light transmitting bricks, Mycelium composite bricks, Bioluminous paints, living bricks for carbon sequestration, Ecobind tiles, CO₂ absorbing concrete Smart Building Materials Introduction – Characteristics of smart materials in comparison to common architectural materials, Types of smart materials and their applications.
Textbooks:	
1.	“Engineering Materials” [Material Science], by Rangwala, Charotar Publications”
2.	“Building Materials” by S.K. Duggal New Age International Publishers.
Reference Books:	
1.	“Engineering Materials”, 5 th edition, by Surindra Singh, Konark Publishers Pvt. Ltd., New Delhi.
2.	“Civil Engg. Materials” by Technical Teachers’ Training Institute, Chandigarh, Tata- McGraw-Hill Publishing Company Ltd., New Delhi.
3.	“Materials of construction”, by R.C. Smith, McGraw-Hill Company, New York.
4.	Engineering Materials”, by Sushil Kumar, Metropolitan Book Co., Pvt. Ltd, New Delhi
5.	Building Construction” Vol.II & III by W.B. McKay, E.L.B.S. and Longman, UK.
6.	Gajanan M. Sabnis, 2016, Green Building With Concrete Sustainable Design and Construction, CRC Press
e-Resources	
1.	https://archive.nptel.ac.in/courses/105/102/105102088/
2.	https://archive.nptel.ac.in/courses/124/105/124105013/
3.	https://archive.nptel.ac.in/courses/105/106/105106206/
4.	https://onlinecourses.nptel.ac.in/noc24_ar20/preview



SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (AUTONOMOUS)

(Approved by AICTE, New Delhi, Affiliated to JNTUK, Kakinada)

Accredited by NAAC with 'A+' Grade.

Recognised as Scientific and Industrial Research Organisation

SRKR MARG, CHINA AMIRAM, BHIMAVARAM – 534204 W.G.Dt., A.P., INDIA

Regulation: R23		II / IV - B.Tech. II - Semester							
CIVIL ENGINEERING									
COURSE STRUCTURE (With effect from 2023-24 admitted Batch onwards)									
Course Code	Course Name	Category	L	T	P	Cr	C.I.E.	S.E.E.	Total Marks
B23HS2201	Managerial Economics and Financial Analysis	HS	2	0	0	2	30	70	100
B23CE2201	Engineering Geology	ES	3	0	0	3	30	70	100
B23CE2202	Concrete Technology	PC	3	0	0	3	30	70	100
B23CE2203	Structural Analysis	PC	3	0	0	3	30	70	100
B23CE2204	Hydraulics & Hydraulic Machinery	PC	3	0	0	3	30	70	100
B23CE2205	Concrete Technology lab	PC	0	0	3	1.5	30	70	100
B23CE2206	Engineering Geology Lab	PC	0	0	3	1.5	30	70	100
B23CE2207	Remote Sensing & Geographical Information Systems	SEC	0	1	2	2	30	70	100
B23CE2208	Design Thinking & Innovation	ES	1	0	2	2	30	70	100
B23MC2202	Environmental Science	MC	2	0	0	-	30	-	30
B23MC2201	English Proficiency	MC	2	0	0	-	30	-	30
TOTAL			19	1	10	21	330	630	960

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
B23HS2201	HS	2	--	--	2	30	70	3Hrs.
MANAGERIAL ECONOMICS AND FINANCIAL ANALYSIS								
(Common to AIDS, CSE, CIC, CSG, CSIT, CE, ECE, EEE, ME)								
Course Objectives: Students are expected to								
1.	Understand the concept and nature of Managerial Economics, its relationship with other disciplines, the Concept of Demand and Demand forecasting							
2.	Familiarize about the Concepts of Cost and Break-Even Analysis							
3.	Learn about accounting cycle and preparation of Financial Statements							
4.	Understand the nature of markets and the Pricing Methods							
5.	Know the concept of Capital, Sources of Raising Finance and Depreciation							
Course Outcomes: At the end of the course the student will be able to								
S.No	Outcome							Knowledge Level
1.	Interpreting the importance of Managerial Economics, demand analysis and methods of demand forecasting							K2
2.	Describe about the usefulness of Cost Analysis and Break Even Analysis							K2
3.	Apply the principles of accounting to convert the transactions and events into Journal, Ledger and Trail balance							K3
4.	Compute the results of Business by preparing Final Accounts							K3
5.	Illustrate the nature of markets and pricing theories							K2
6.	Explain the Types of capital, their sources and importance & estimation of Depreciation							K2
SYLLABUS								
UNIT-I (12 Hrs)	Introduction to Managerial Economics and demand Analysis: Managerial Economics: Definition of Economics & Classification of Economics (Micro & Macro), Meaning, Nature, & Scope of Managerial Economics. Demand Analysis: Concept of Demand, Determinants of Demand, Demand schedule, Demand curve, Law of Demand and its exceptions. Elasticity of Demand, Types of Elasticity of Demand. Importance of Demand Forecasting and its Methods.							
UNIT-II (8 Hrs)	Cost Analysis: Importance of cost analysis, Types of Cost- Actual cost Vs Opportunity cost, Fixed cost Vs Variable cost, Explicit Vs Implicit cost, Historical cost Vs Replacement cost, Incremental cost Vs Sunk cost; Elements of costs – Material, Labour, Expenses; Methods of costing - Job costing, contract costing, Process costing, Batch costing, Unit							

	costing, Service costing, Multiple costing. Break-even analysis: Determination of Break-even Point Applications, Assumptions and Limitations of Break-even analysis (Theory only).
UNIT-III (12 Hrs)	Introduction to Financial Accounting: Importance of Accounting - Double Entry System of Accounting - Types of Accounts - Journal, Ledger, Trail Balance, Trading Account, Profit and Loss Account and Balance Sheet (Final Accounts with Simple adjustments).
UNIT-IV (8 Hrs)	Introduction to Markets & Pricing Policies Market Structures: Salient Features of Perfect Competition, Monopoly, Monopolistic competition, Oligopoly and Duopoly. Pricing: Importance of pricing and its meaning; Methods of Pricing: Cost Based -Full cost, Mark-up, Marginal & Break-even; Demand Based -Penetrating, Skimming; Competition Based -Going rate, Sealed Bid, Discount; Internet Pricing -Flat-rate, Usage sensitive.
UNIT-V (8 Hrs)	Capital & Depreciation: Types of Capital-Fixed capital & Working Capital, Components of Working Capital, Factors influencing Working capital. Methods of Raising Finance - Short term, Medium term and Long term. Depreciation – Meaning, Importance and causes of depreciation; Methods of Depreciation-Straight line and Diminishing Balancing methods (Theory only).
Text Books:	
1.	AR Aryasri, Managerial Economics and Financial Analysis, TMH Pvt. Ltd, New Delhi
2.	Dr. N. Appa Rao, Dr. P. Vijaya kumar: Managerial Economics and Financial Analysis, Cengage Publications, NewDelhi
Reference Books:	
1.	Dr.B.Kuberudu & T.V.Ramana :Managerial Economics and Financial anaysis, Himalaya Publishing House
2.	Varshney R.L, K.L Maheswari, Managerial Economics, S. Chand & Company Ltd,
3.	Shashi K.Gupta & R.K.Sharma Management Accounting, Kalyani Publishers
4.	Maheswari S.N, An Introduction to Accountancy, Vikas Publishing House Pvt Ltd

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
B23CE2201	ES	3	--	--	3	30	70	3 Hrs.
ENGINEERING GEOLOGY								
(For CE)								
Course Objectives: The objectives of this course is to make the student aware of								
1.	Geological agents and their work.							
2.	Petrological and Mineralogical concepts							
3.	Geological Structures (Joint, veins, crack, faults, and fold), reasons of formation for each type and their effects on the engineering projects							
4.	Ground water, occurrence-exploration methods. Natural hazards and preventive/control methods							
5.	Application of geology in Civil Engineering works viz. Dams and Tunnels.							
Course Outcomes: At the end of the course students will be able to								
S.No	Outcome							Knowledge Level
1.	Describe earth-atmospheric processes and their impact on civil engineering works.							K2
2.	Summarize characteristics of minerals and rocks.							K2
3.	Explain structural geological conditions of the site.							K2
4.	Illustrate geophysical methods and its use in site investigations							K2
5.	Interpret site suitability for engineering project on the basis of geological conditions							K2
SYLLABUS								
UNIT-I (8 Hrs)	Introduction: Branches of Geology, Importance of Geology in Civil Engineering with case studies. Weathering: Weathering of rocks, Geological agents, Geological work of rivers							
UNIT-II (10 Hrs)	Mineralogy and Petrology: Definitions of mineral and Rock-Different methods of study of mineral and rock. Physical properties of minerals and rocks for Megascopic study for the following minerals and rocks. Common rock forming minerals: Feldspar, Quartz Group, Olivine, Augite, Hornblende, Mica Group, Asbestos, Talc, Chlorite, Kyanite, Garnet, Calcite. Ore forming minerals: Pyrite, Hematite, Magnetite, Chlorite, Galena, Pyrolusite, Graphite, Chromite, Magnetite and Bauxite. Classification, structures, textures and forms of Igneous rocks, Sedimentary rocks, Metamorphic rocks, Megascopic study of granite varieties (pink, gray, green), Pegmatite, Dolerite, Basalt, Shale, Sand Stone, Lime Stone, Laterite, Quartzite, Gneiss, Schist,							

	Marble, Khondalite and Slate.Engineering properties of rocks.
UNIT-III (8 Hrs)	Structural Geology: Strike, Dip and Outcrop, Study of common geological structures associating with the rocks - Folds, Faults, Joints and Unconformities- parts, types, mechanism and their importance in Civil Engineering
UNIT-IV (12 Hrs)	Ground Water: Water table, Cone of depression, Geological controls of Ground Water Movement, Ground Water Exploration Techniques. Earthquakes and Land Slides: Terminology, Classification, causes and effects, Shield areas and Seismic belts, Richter scale intensity, Precautions of building constructions in seismic areas. Classification of Landslides, Causes and Effects, measures to be taken prevent their occurrence at Landslides. Geophysics: Importance of Geophysical methods, Classification, Principles of Geophysical study: Gravity method, Magnetic method, Electrical methods, Seismic methods, Radiometric method and Electrical resistivity, Seismic refraction methods.
UNIT-V (8 Hrs)	Geological Investigations of Dams, Reservoirs and Tunnels: Types of Dams, Geological considerations in the selection of a Dam site. Geology consideration for successful constructions of reservoirs, Life of Reservoirs. Tunneling-types, purpose, effects, Lining of Tunnels. Influence of Geology for successful Tunneling.
Textbooks:	
1.	Engineering and General Geology by Parbin Singh – Katson Publishing House.
2.	Engineering Geology by N.Chennakesavulu, Mc-Millan, India Ltd. 2005.
Reference Books:	
1.	Principles of Engineering Geology by KVGK Gokhale. B.S Publications-2005.
2.	Principles and Applications of Photo Geology by Pandey, Willey Eastern Limited.
3.	Engineering Geology by Subinoy Gangopadhyay, Oxford University Press.
4.	Fundamentals of Engineering Geology by P.G. Bell, B.S.P. Publications, 2012.
5.	Engineering Geology and Geo-techniques by F.G. Bell.
e-Resources	
1.	https://onlinecourses.nptel.ac.in/noc23_ce107/preview
2.	https://nptel.ac.in/courses/105105106

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
B23CE2202	PC	3	--	--	3	30	70	3 Hrs.
CONCRETE TECHNOLOGY								
(For CE)								
Course Objectives: The objectives of this course is to make the student aware of								
1.	Cement composition, hydration processes, and types, including recent advancements in concrete materials.							
2.	Concrete production techniques, properties, and testing methodologies comprehensively.							
3.	Recent advancements in concrete technology into practical applications, exploring innovative materials and techniques that enhance sustainability and performance in construction.							
Course Outcomes: At the end of the course students will be able to								
S.No	Outcome							Knowledge Level
1.	Use appropriate materials for enhancing the performance of concrete.							K3
2.	Predict the behavior of fresh concrete by conducting workability tests.							K3
3.	Apply destructive and non-destructive testing methods for determining properties of hardened concrete.							K3
4.	Predict long-term behaviour of concrete by understanding the effect of various factors influencing it.							K3
5.	Calculate concrete mix proportions by using IS and ACI standards and select suitable special concrete based on real time situations.							K3
SYLLABUS								
UNIT-I (10Hrs)	Materials For Concrete: Cement: Raw materials; composition - Hydration, chemistry of cement; Admixtures: Types of Admixtures- super plasticizers, plasticizers, retarders, accelerators, air entrained admixtures and pozzolanic admixtures – fly ash and silica fume Aggregates: Source- natural and artificial, alkali aggregate reaction. Sieve analysis – Fineness modulus – Grading curves – Grading of fine & coarse Aggregates – Maximum aggregate size- Quality of mixing water							
UNIT-II (8 Hrs)	Fresh Concrete: Steps in Manufacture of Concrete–Batching of ingredients; mixing, transporting, and placing - Compacting, finishing, and curing of concrete - Workability, bleeding and segregation of concrete - Factors influencing it - Tests on workability of concrete, Ready mix Concrete, Shotcrete.							

UNIT-III (10 Hrs)	Hardened Concrete: Water / Cement ratio – Abram’s Law – Gel/space ratio – Nature of strength of concrete –Maturity concept – Compressive strength and parameters affecting it - Tensile strength - direct and indirect; Modulus of elasticity and Poisson's ratio, flexural strength of concrete- Non-destructive testing methods – Codal provisions for NDT.
UNIT-IV (8Hrs)	Elasticity, Creep & Shrinkage – Modulus of elasticity – Dynamic modulus of elasticity – Poisson’s ratio – Creep of concrete – Factors influencing creep – Relation between creep & time – Nature of creep – Effects of creep – Shrinkage –types of shrinkage.
UNIT-V (10 Hrs)	Mix Design And Special Concretes: Properties and applications of: fibre reinforced concrete, high strength concrete, high performance concrete, lightweight concrete and self-compacting concrete, Polymer concrete, geo polymer concrete . Concept of Mix proportion - Quality acceptance criteria as per Indian standard method. Design of concrete mixes as per IS-10262 and ACI method.
Textbooks:	
1.	Concrete Technology by M. S. Shetty. – S. Chand & Co.; 2004
2.	Properties of Concrete by A.M. Neville – PEARSON – 4th edition
Reference Books:	
1.	Concrete Technology by M.L. Gambhir. – Tata Mc.Graw Hill Publishers, New Delhi 5 th edition 2013
2.	Concrete Technology by Job Thomas, Cengage Publications, 1 st edition, 2015
3.	Concrete Microstructure, Properties of Materials by P.K. Mehta and Moterio. McGraw Hill 4 th edition 2014
4.	Concrete Technology, J.J. Brooks and A. M. Neville, Pearson, 2019, 2nd Edition
5.	Concrete Technology by A.R. Santha Kumar, Oxford University Press, New Delhi
e-Resources	
1.	https://archive.nptel.ac.in/courses/105/102/105102012/
2.	https://archive.nptel.ac.in/courses/105/106/105106206/

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
B23CE2203	PC	3	--	--	3	30	70	3 Hrs.
STRUCTURAL ANALYSIS								
(For CE)								
Course Objectives: The objectives of this course is to make the student aware of								
1.	Deflection of simple determinate beams, frames & trusses							
2.	Method of consistent deformation and energy methods.							
3.	Indeterminate continuous beams and Rigid frames by force and displacement methods.							
4.	The influence line diagrams and moving loads for statically determinate beams.							
Course Outcomes: At the end of the course students will be able to								
S.No	Outcome							Knowledge Level
1.	Calculate the deformations of the determinate beams, rigid frames and trusses by Unit load method and for beams by geometrical method.							K3
2.	Compute the member end moments of statically indeterminate single span and two span beams subjected to different loads using law of Reciprocal Deflection and theorem of least work.							K3
3.	Solve the member end moments for continuous beams and for rigid frames by method of consistent deformation.							K3
4.	Determine the member end moments for continuous beams and rigid frames by slope deflection method.							K3
5.	Find reactions, BM and SF in beams subjected to moving loads using ILD							K3
SYLLABUS								
UNIT-I (12 Hrs)	Deformation of statically determinate beams by using (i) Moment area method (ii) Conjugate beam method (iii) Unit load method Strain- energy due to (i) Axial load (ii) Bending Moment (iii) Shear force and (iv) Torque Deformation of Statically Determinate rigid frame and trusses: (i) Determination of slopes and deflection of statically determinate rigid frames using Unit load method (ii) Determination of deflection of statically determinate trusses using Unit load method							
UNIT-II (8 Hrs)	Analysis of Statically indeterminate beam a) Analysis of propped cantilever beams and fixed beams subjected to concentrated load and uniform distributed load by method of consistent deformation. b) Analysis of two span continuous beams by using law of Reciprocal Deflection and theorem of least work and induced reaction due to yielding of supports.							
UNIT-III	Analysis of Statically indeterminate continuous beam and rigid frames by force							

(12 Hrs)	method. a) Analysis of three span continuous beams due to applied loads by theorem of three moments b) Analysis of three span continuous beams due to uneven support settlements by theorem of three moments c) Analysis of statically indeterminate rigid frames due to applied loads by method of consistent deformation. d) Analysis of induced reactions on statically indeterminate rigid frames due to yielding of supports by method of consistent deformation.
UNIT-IV (10 Hrs)	Analysis of Statically indeterminate continuous beam and rigid frames by Displacement Method. a) Analysis of statically indeterminate three span continuous beams due to applied load by slope deflection method. b) Analysis of statically indeterminate three span continuous beams for uneven support settlements by slope deflection method. c) Analysis of statically indeterminate rigid frames (without side sway and with side sway) by slope deflection method. d) Analysis of statically indeterminate rigid frames due to yielding of supports by slope deflection method.
UNIT-V (8 Hrs)	Influence Lines Definition – Influence line for Reaction, SF and BM in case of simply supported and overhanging beams-Load position for Max SF at a section –Load position for max BM at a section- Single point load, U.D.L longer than the span, U.D.L. shorter than the span. Moving loads – Max SF and BM at a given section and absolute MaxSF and BM due to single concentrated load, U.D.L. longer than the span, U.D.L. shorter than the span, two point loads with fixed distance between them and several point loads, Maximum Bending moment at a section under a wheel load and absolute maximum Bending moment in the case of several wheel loads-Equivalent uniformly Distributed load, Focal length.
Textbooks:	
1.	Basic Structural Analysis, C.S. Reddy, Mc Graw Hill Education (India) PVT.LTD
2.	Structural Analysis- T.S. Thandavamoorthy, Oxford University Press, New Delhi
Reference Books:	
1.	Statically indeterminate structures – C.K. Wang, Mc Graw Hill Education PVT.LTD
2.	Mechanics of structures Vol. II- S.B.Junnarkar and Dr.H.J.shah, Charotar Publishing House.
3.	Structural analysis – Devdas Menon, Narosa Publishing House PVT.LTD
4.	Theory of Structures Volume II, S.P Gupta, G.S Pandit, R. Gupta – Tata McGraw-Hill Publishing Company Limited, New Delhi
e-Resources	
1.	https://archive.nptel.ac.in/courses/105/101/105101085/
2.	https://archive.nptel.ac.in/courses/105/105/105105166/

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
B23CE2204	PC	3	--	--	3	30	70	3 Hrs.
HYDRAULICS AND HYDRAULIC MACHINERY								
(For CE)								
Course Objectives: The objectives of this course is to make the student aware of								
1.	Laminar and turbulent flow in pipes, dimensional analysis and boundary layer theory.							
2.	Analysis of uniform and non-uniform flows through the open channels.							
3.	Analysis and design of turbines and pumps							
Course Outcomes: At the end of the course students will be able to								
S.No	Outcome							Knowledge Level
1.	Solve the characteristics of laminar and turbulent flow in pipes, the physical phenomenon of a fluid problem and boundary layer theory.							K3
2.	Calculate geometrical parameters of open channels to address the uniform flow problems.							K3
3.	Solve behavior of non-uniform flow in open channels.							K3
4.	Compute the performance of the impact of jets on plates and analysis and design of turbines.							K3
5.	Determine the performance of the pumps.							K3
SYLLABUS								
UNIT-I (10Hrs)	Laminar & Turbulent flow in pipes: Laminar Flow- Laminar flow through: circular pipes, annulus and parallel plates. Stoke's law, Measurement of viscosity. Reynolds experiment, Transition from laminar to turbulent flow. Resistance to fluid flow in smooth and rough pipes, Moody's diagram – Dimensional analysis: dimensional analysis using Rayleigh's and Buckingham's π - π -methods, Model laws. Introduction to Boundary Layer Theory							
UNIT-II (10 Hrs)	Uniform flow in Open Channels: Open Channel Flow - Comparison between open channel flow and pipe flow, geometrical parameters of a channel, classification of open channels, classification of open channel flow, Velocity Distribution of channel section. Hydraulically efficient channel sections: Rectangular, trapezoidal and triangular channels, Energy and Momentum correction factors							
UNIT-III (10 Hrs)	Non-uniform flow in Open Channels: Specific energy, critical flow, discharge curve, Specific force, Specific depth, and Critical depth. Measurement of Discharge and Velocity – Gradually Varied Flow- Dynamic Equation of Gradually Varied Flow. Hydraulic Jump and classification - Elements and characteristics- Energy dissipation							

UNIT-IV (10 Hrs)	Impact of Jets: Hydrodynamic force of jets on stationary and moving flat, inclined and curved vanes - Velocity triangles at inlet and outlet - Work done and efficiency Hydraulic Turbines: Classification of turbines; Pelton wheel and its design. Francis's turbine and its design - efficiency - Draft tube: theory - characteristic curves of hydraulic turbines. Cavitation: causes and effects.
UNIT-V (10 Hrs)	Pumps: Working principles of a centrifugal pump, work done by impeller; heads, losses and efficiencies; minimum starting speed; Priming; specific speed; limitation of suction lift, net positive suction head (NPSH); Performance and characteristic curves; Cavitation effects; Multistage centrifugal pumps; troubles and remedies
Textbooks:	
1.	P. M. Modi and S. M. Seth, Hydraulics and Fluid Mechanics, Standard Book House 22 nd , 2019
2.	K. Subrahmanya, Theory and Applications of Fluid Mechanics, Tata McGraw Hill, 2 nd edition 2018
Reference Books:	
1.	R. K. Bansal, A text of Fluid mechanics and hydraulic machines, Laxmi Publications (P) Ltd., New Delhi 11 th edition, 2024
2.	Fluid Mechanics by Frank M. White, Henry Xue, Tata McGraw Hill, 9 th edition, 2022
3.	C. S. P. Ojha, R. Berndtsson and P. N. Chadramouli, Fluid Mechanics and Machinery, Oxford University Press, 2010
4.	Introduction to Fluid Mechanics & Fluid Machines by S K Som, Gautam Biswas, S Chakraborty 3 rd edition 2011
e-Resources	
1.	https://archive.nptel.ac.in/courses/105/106/105106114/
2.	https://nptel.ac.in/courses/105103021

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
B23CE2205	PC	--	--	3	1.5	30	70	3 Hrs.
CONCRETE TECHNOLOGY LAB								
(For CE)								
Course Objectives:								
The objective of this course is to familiarize students with testing the fundamental properties of both fresh and hardened concrete ingredients.								
Course Outcomes: At the end of this course student will be able to								
S. No	Outcome							Knowledge Level
1	Outline importance of testing cement and its properties							K3
2	Assess different properties of Aggregates							K3
3	Assess fresh concrete properties and their relevance to hardened concrete							K3
4	Assess hardened concrete properties							K3
SYLLABUS								
Students have to perform any 10 of the following Experiments:								
1.	Tests on Cement							
	a. Normal Consistency and Fineness of cement.							
	b. Initial setting time and Final setting time of cement.							
	c. Specific gravity and soundness of cement.							
	d. Compressive strength of cement.							
2.	Tests on Fine Aggregates and Coarse Aggregates							
	a. Grading and fineness modulus of Fine aggregate by sieve analysis.							
	b. Specific gravity of fine and Coarse aggregate							
	c. Water absorption and Bulking of sand							
3.	Tests on fresh Concrete							
	a. Workability of concrete by compaction factor test.							
	b. Workability of concrete by slump test.							
	c. Workability of concrete by Vee-bee test.							
4.	Tests on Hardened Concrete							
	a. Compressive strength of cement concrete and Modulus of rupture							
	b. Young's Modulus and Poisson's Ratio							
	c. Split tensile strength of concrete.							
	d. Non-Destructive testing on concrete (for demonstration)							
Reference Books:								
1	Concrete Technology by M.L.Gambhir. –TataMc. Graw Hill publishers, NewDelhi.							
2	Concrete Technology by M.S.Shetty. –S.Chand&Co.;2004							

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
B23CE2206	PC	--	--	3	1.5	30	70	3 Hrs.
ENGINEERING GEOLOGY LAB								
(For CE)								
Course Objectives: The objective of this course is to make the student aware of								
1	Physical properties of minerals							
2	Megascopic characteristics of rocks							
3	Geomorphological process and their importance							
4	Structural geology and Stratigraphy significance in engineering project models							
Course Outcomes: At the end of this course student will be able to								
S.No	Outcome							Knowledge Level
1	Determine minerals using physical properties							K3
2	Find the rocks according to mega-scopic properties							K3
3	Demonstrate geomorphological and structural geology models							K3
4	Infer the terrain slope by using strike-dip problems							K4
SYLLABUS								
1	Study of physical properties and identification of minerals Rock forming minerals – Quartz group, Feldspar group, Mica group & Talc, Chlorite, Olivine, Kyanite, Asbestos, Calcite, Gypsum. Ore forming minerals – Magnetite, Hematite, Pyrite, Graphite.							
2	Megascopic identification of rocks and their Engineering properties Igneous rocks – Types of Granite, Pegmatite, Gabbro, Dolerite, Syenite, Granite Poryphery, Basalt. Sedimentary rocks – Sandstone, Ferruginous sandstone, Lime stone, Shale, Laterite, Conglomerate. Metamorphic rocks – Gneiss, Slate, Schist, Marble, Khondalite, Quartzite.							
3	Description and Identification of Geomorphological models-River, Ground water, Sea waves and currents							
4	Description and Identification of Structural geology models- Folds, Faults							
5	Simple Structural Geology problems on Dip and Strike							
6	Field work – To identify Minerals, Rocks, Geomorphology& Structural Geology							
Reference Books:								
1	Applied Engineering Geology Practicals by M T Mauthesha Reddy, New Age International Publishers, 2nd Edition.							
2	Foundations of Engineering Geology by Tony Waltham, Spon Press, 3rd Edition, 2009.							

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
B23CE2207	SEC	--	1	2	2	30	70	3 Hrs.
REMOTE SENSING AND GEOGRAPHICAL INFORMATION SYSTEMS								
(For CE)								
Course Objectives: The objective of this course is to make the student aware of								
1	Concept of Remote Sensing phenomena and its components							
2	Interpretation of satellite data							
3	Familiar with Geographical Information System							
4	Geospatial analysis							
Course Outcomes: After completion of this course, students will be able to								
S.No	Outcome							Knowledge Level
1	Describe the basic concepts of Remote sensing							K2
2	Interpret the various types of satellite data.							K2
3	Summarize the concepts of GIS and its applications.							K2
4	Compute the geometry of physical features using GIS software							K3
5	Determine hydrological characteristics of a site using geospatial analysis							K3
SYLLABUS								
Unit-I	Introduction to Remote sensing: History of Remote Sensing, Electromagnetic Radiation, Electromagnetic Spectrum, Energy Interaction with Atmosphere, Energy Interaction with the Earth Surfaces - Characteristics of Remote Sensing Systems, Sensor Resolutions, Advantages & Limitations - Platforms: Types of Sensors, Airborne Remote Sensing, Space borne Remote Sensing - IRS, LANDSAT, SPOT & Recent satellite.							
Unit-II	Digital Image analysis: Digital Image Characteristics, Digital Image Data Formats, Band Interleaved by Pixel (BIP), Band Interleaved by Line (BIL), Band Sequential (BSQ) - Visual Interpretation Elements, Preprocessing, Enhancement, Classification, Supervised classification, Unsupervised classification.							
Unit-III	Introduction to Geographic Information System: Principles, Components and Applications of GIS - Map projections, Spatial Data Structures, Raster and Vector Data Formats, Data Inputs, Data Manipulation, Data Retrieval, Data Analysis - Spatial data analysis: Overlay Function-Vector Overlay Operations, Raster Overlay Operations, Arithmetic Operators, Comparison and Logical Operators, Conditional Expressions - Network Analysis: Components of network, Transportation network - Optimum path analysis							

EXERCISES	
1	Visual Interpretation of standard FCC
2	Georeferencing a Toposheet or Map
3	Digitization of physical features on a map/image using GIS software.
4	Creation of Thematic Map
5	Calculation of Feature geometry – Length, Area and Perimeter
6	Developing Digital Elevation Model
7	Stream network – Stream ordering map.
8	Watershed - Calculate Hydro-geomorphological parameters
9	Transportation Network Map – Route analysis
Reference Books:	
1	Basudeb Bhatta (2021). 'Remote sensing and GIS', 3 rd edn., Oxford University Press
2	S. Kumar, (2016) 'Basics of Remote sensing & GIS', Laxmi Publications
3	QGIS User Guide ArcGIS User Manual by ESRI



Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
B23CE2208	ES	1	--	2	2	30	70	3 Hrs.
DESIGN THINKING & INNOVATION								
(Common to all Programmes of Engineering)								
Course Objectives:								
1.	Bring awareness on innovative design and new product development.							
2.	Explain the basics of design thinking.							
3.	Familiarize the role of reverse engineering in product development.							
4.	Train how to identify the needs of society and convert into demand.							
5.	Introduce product planning and product development process.							
Course Outcomes: After completion of this course, students will be able to								
S.No	Outcome							Knowledge Level
1.	Define the concepts related to design thinking.							K1
2.	Explain the fundamentals of Design Thinking and innovation.							K2
3.	Apply the design thinking techniques for solving problems in various sectors.							K3
4.	Analyse to work in a multidisciplinary environment.							K4
5.	Evaluate the value of creativity.							K5
SYLLABUS								
UNIT-I (10Hrs)	Introduction to elements and principles of Design, basics of design-dot, line, shape, form as fundamental design components. Principles of design. Introduction to design thinking, history of Design Thinking, New materials in Industry.							
UNIT-II (10 Hrs)	Design thinking process (empathize, analyze, idea & prototype), implementing the process in driving inventions, design thinking in social innovations. Tools of design thinking - person, costumer, journey map, brainstorming, product development. Activity: Every student presents their idea in three minutes, Every student can present design process in the form of flow diagram or flow chart etc. Every student should explain about product development.							
UNIT-III (10 Hrs)	Art of innovation, Difference between innovation and creativity, role of creativity and innovation in organizations. Creativity to Innovation. Teams for innovation, Measuring the impact and value of creativity. Activity: Debate on innovation and creativity, Flow and planning from idea to innovation, Debate on value-based innovation.							
UNIT-IV (10 Hrs)	Problem formation, introduction to product design, Product strategies, Product value, Product planning, product specifications. Innovation towards product design Case studies.							

	Activity: Importance of modeling, how to set specifications, Explaining their own product design.
UNIT-V (10 Hrs)	Design Thinking applied in Business & Strategic Innovation, Design Thinking principles that redefine business – Business challenges: Growth, Predictability, Change, Maintaining Relevance, Extreme competition, Standardization. Design thinking to meet corporate needs. Design thinking for Startups. Defining and testing Business Models and Business Cases. Developing & testing prototypes. Activity: How to market our own product, about maintenance, Reliability and plan for startup.
Textbooks:	
1.	Tim Brown, Change by design, 1/e, Harper Bollins, 2009.
2.	Idris Mootee, Design Thinking for Strategic Innovation, 1/e, Adams Media, 2014.
Reference Books:	
1.	David Lee, Design Thinking in the Classroom, Ulysses press, 2018.
2.	Shrrutin N Shetty, Design the Future, 1/e, Norton Press, 2018.
3.	William lidwell, Kritinaholden, &Jill butter, Universal principles of design, 2/e, Rockport Publishers, 2010.
4.	Chesbrough.H, The era of open innovation, 2003.
e-Resources:	
1.	https://nptel.ac.in/courses/110/106/110106124/
2.	https://nptel.ac.in/courses/109/104/109104109/
3.	https://swayam.gov.in/nd1_noc19_mg60/preview
4.	https://onlinecourses.nptel.ac.in/noc22_de16/preview

Course Code	Category	L	T	P	C	C.I.E	S.E.E	Exam
B23MC2202	MC	2	--	--	--	30	--	--
ENVIRONMENTAL SCIENCE								
(Common to AIDS, AIML, CE, CSBS, CSIT, IT and ME.)								
Course Objectives: The objective of the course is to impart:								
1.	Overall view on natural resources.							
2.	Awareness on ecosystem and its services.							
3.	Various environmental challenges induced due to unplanned anthropogenic activities.							
4.	Consciousness on the social issues, environmental legislation and global treaties							
Course Outcomes: At the end of the course, the students will be able to								
S. No	Outcome							Knowledge Level
1	Describe natural resources and their interaction							K2
2	Illustrate ecosystem types, biodiversity and conservation strategies							K2
3	Summarize contaminants of environment and preventive methods							K2
4	Explain protection of environment by employing constitutional provisions							K2
5	Explain global scenario of surroundings and social conditions							K2
SYLLABUS								
UNIT-I (8 Hrs)	Multidisciplinary Nature of Environmental Studies: Definition, Scope and Importance – Need for Public Awareness. Natural Resources: Renewable and non-renewable resources – Natural resources and associated problems. Forest resources – Use and over – exploitation, deforestation, case studies – Timber extraction – Mining, dams and other effects on forest and tribal people. Water resources – Use and over utilization of surface and ground water – Floods, drought, conflicts over water, dams – benefits and problems. Mineral resources: Use and exploitation, environmental effects of extracting and using mineral resources, case studies. Food resources: World food problems, changes caused by agriculture and overgrazing, effects of modern agriculture, fertilizer-pesticide problems, water logging, salinity, case studies. Energy resources: Growing energy needs, renewable and non-renewable energy sources use of alternate energy sources							
UNIT-II (8 Hrs)	Ecosystems: Concept of an ecosystem. – Structure and function of an ecosystem – Producers, consumers and decomposers – Energy flow in the ecosystem – Ecological succession – Food chains, food webs and ecological pyramids – Introduction, types, characteristic features, structure and function of the following ecosystem: a. Forest ecosystem. b. Grassland ecosystem c. Desert ecosystem d. Aquatic ecosystems (ponds, streams, lakes, rivers, oceans, estuaries) Biodiversity and Its Conservation : Introduction and Definition - genetic, species and							

	ecosystem diversity – Bio-geographical classification of India – Value of biodiversity: consumptive use, Productive use, social, ethical, aesthetic and option values – Biodiversity at global, National and local levels – India as a mega-diversity nation – Hot-spots of biodiversity – Threats to biodiversity: habitat loss, poaching of wildlife, man-wildlife conflicts – Endangered and endemic species of India – Conservation of biodiversity: In-situ and Ex-situ conservation of biodiversity.
UNIT-III (6 Hrs)	Environmental Pollution: Definition, Cause, effects and control measures of: a. Air Pollution. b. Water pollution c. Soil pollution d. Marine pollution e. Noise pollution f. Thermal pollution g. Nuclear hazards. Solid Waste Management: Causes, effects and control measures of urban and industrial wastes – Role of an individual in prevention of pollution – Pollution case studies – Disaster management: floods, earthquake, cyclone and landslides
UNIT-IV (6 Hrs)	Social Issues and the Environment: From Unsustainable to Sustainable development – Urban problems related to energy watershed management – Resettlement and rehabilitation of people; its problems and concerns – Carbon credits, Mission LiFE. Environmental ethics: Issues and possible solutions. Climate change, global warming, acid rain, ozone layer depletion, nuclear accidents and holocaust –Wasteland reclamation – Consumerism and waste products. Environment Protection Act. – Air (Prevention and Control of Pollution) Act. – Water (Prevention and control of Pollution) Act – Wildlife Protection Act – Forest Conservation Act. Issues involved in enforcement of environmental legislation – Public awareness.
UNIT-V (6 Hrs)	Human Population And The Environment: Population growth, variation among nations. Population explosion – Family Welfare Programmes. – Environment and human health – Human Rights – Value Education – HIV/AIDS – Women and Child Welfare – Role of information Technology in Environment and human health – Case studies. Field Work: Visit to a local area to document environmental assets River/forest grassland/hill/mountain – Visit to a local polluted site-Urban/Rural/Industrial/Agricultural Study of common plants, insects, and birds – river, hill slopes, etc.
Text Books:	
1.	Erach Bharucha, Text book of Environmental Studies for Undergraduate Courses, Universities Press (India) Private Limited, 2019.
2.	Palaniswamy, Environmental Studies, 2/e, Pearson education, 2014.
3.	S. Azeem Unnisa, Environmental Studies, Academic Publishing Company, 2021.
4.	K. Raghavan Nambiar, “Text book of Environmental Studies for Undergraduate Courses as per UGC model syllabus”, SciTech Publications (India), Pvt. Ltd, 2010.
5	K. V. S. G. Murali Krishna, The Book of Environmental Studies, Savera Publishing House.
6	Environmental Studies, R. Rajagopalan, 2 nd Edition, 2011, Oxford University Press.

Reference Books:	
1.	Deeksha Dave and S.S. Katewa, Textbook of Environmental Studies, 2/e, Cengage Learning.
2.	M. Anji Reddy, “Textbook of Environmental Sciences and Technology”, BS Publication, 2014.
3.	J.P. Sharma, Comprehensive Environmental studies, Laxmi publications, 2006.
4.	J. Glynn Henry and Gary W. Heinke, Environmental Sciences and Engineering, Prentice Hall of India Private limited, 1988.
5.	G.R. Chatwal, A Text Book of Environmental Studies, Himalaya Publishing House, 2018
6.	Gilbert M. Masters and Wendell P. Ela, Introduction to Environmental Engineering and Science, 1/e, Prentice Hall of India Private limited, 1991.
e-Resources	
1.	https://onlinecourses.nptel.ac.in/noc23_hs155/preview
2.	https://www.edx.org/learn/environmental-science/rice-university-ap-r-environmental-science-part-3-pollution-and-resources?index=product&objectID=course-3a6da9f2-lec07.pdf (iasri.res.in)
3.	http://ecoursesonline.iasri.res.in/Courses/Environmental%20Science-I/Data%20Files/pdf/lec07.pdf
4.	https://www.youtube.com/watch?v=5QxxaVfgQ3k



Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
B23MC2201	MC	2	--	--	--	30	--	--
ENGLISH PROFICIENCY								
(Common to CSE, CE, ECE, EEE, CIC, CSD).								
Course Outcomes: After completion of this course, students will be able to								
S No	Outcome							Knowledge Level
1.	Interpret IELTS & TOEFL listening comprehension texts.							K2
2.	Demonstrate essential speaking skills in academic, professional, and real-life contexts.							K2
3.	Interpret the written discourse by applying effective reading strategies.							K2
4.	Construct coherent and cohesive paragraphs, e-mails, letters, and essays.							K3
SYLLABUS								
UNIT-I	Listening Skills Listening as a receptive skill Listening skills for IELTS, TOEFL, Duolingo, PTE, and other competitive examinations.							
UNIT-II	Speaking Skills JAM/ Extempore Debate / Group Discussion Presentation Skills							
UNIT-III	Reading Skills Types of Reading (Intensive and Extensive Reading, Skimming, Scanning) Reading/Summarizing News Paper Articles							
UNIT-IV	Writing Skills Essay Writing (Argumentative, Analytical and Descriptive essays for GRE, TOEFL & IELTS) E-Mail / Letter Writing Resume Writing							
UNIT-V	Integrated Language Skills Listening Skills for Speaking and Writing Reading Skills for Writing and Speaking							
Text Books:								
1.	Interchange (5 th edition) by Jack C.Richards, CUP, 2017.							

Reference Books:	
1.	Fundamentals of Technical Communication (1 st edition) by Meenakshi Raman, Sangeeta Sharma of OUP, 2014.
2.	The Oxford Guide to Writing and Speaking (3 rd edition) by John Seely OUP, 2013.
3.	Effective Technical Communication (2 nd edition) by M. Ashraf Rizwi. TataMcGrawhill, 2017.
e-Resources:	
1	BBC Learning English - Learn English with BBC Learning English - Homepage
2	Grammar Learn English (britishcouncil.org)
3	Duolingo English Test
4	IELTS Test Preparation Materials - Videos, Practice tests, Articles and More (idp.com)

